Chemistry Exam 14 — Chapters 1–9 (Full Syllabus)

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11/12	31.5/33	17.5/20	60.0/65	A+
SECTION A (MCQS)	SECTION B (SHORT)	SECTION C (LONG)	TOTAL — 92.3%	GRADE

Section A — Multiple Choice Questions (11/12)

Q	Question Summary	Correct	Seemab	Result	Marks
1	Branch that studies heat changes and reaction rates	A — Physical chemistry	A	✓	1
2	Direct change of gas to solid	B — Deposition	B	✓	1
3	Maximum electrons in a *d* sub-shell	C — 10	C	✓	1
4	Electronic configuration of Cl^- (Z=17)	D — 2, 8, 8	D	✓	1
5	Atom with the largest atomic size	C — K	C	✓	1
6	Inert element with a completely filled outer shell	D — Argon	D	✓	1
7	Substance containing a metallic bond	A — Copper	A	✓	1
8	Mass of 0.25 mol of $CaCO_3$	B — 25 g	B	✓	1
9	Oxygen atoms in one mole of CO_2	D — 1.204×10^{24}	D	✓	1
10	Oxidation state of Cl in ClO_3^-	A — +5	A	✓	1
11	Minimum energy to start a reaction	B — Activation energy	B	✓	1
12	A reversible reaction reaches equilibrium only if carried out in…	C — a closed system	D	✗	0

MCQ 11/12. The single miss is Q12: she chose D (“conditions of high pressure only”); equilibrium requires a **closed system** (C) so nothing can escape. This is Chemistry’s first-ever MCQ error — the subject was 72/72 across the previous six papers — and it ends the 13-exam consecutive perfect-MCQ streak (last slip was Math exam-07 on 5 Jun).

Section B — Short Questions (31.5/33)

2(i) — MAIN Define analytical & physical chemistry; one thing each studies — 3/3

Analytical chemistry defined as the branch dealing with the methods and instruments for determining the composition and structure of matter ✓. Physical chemistry defined as the branch dealing with laws and theories to understand the changes and structure of matter ✓. Both definitions acceptable; full marks.

2(ii) — MAIN Differentiate homogeneous vs heterogeneous mixture, with examples — 3/3

Clean two-column table: uniform vs non-uniform composition; single vs two-or-more phases; components invisible vs visible to the eye. Examples salt-in-water / air and sand-in-water / salad. Three valid differences and two examples each — full marks.

2(iii) — MAIN Maximum electrons in K, L, M shells and the rule — 3/3

Rule $2n^2$ stated ✓. K = $2(1)^2$ = 2, L = $2(2)^2$ = 8, M = $2(3)^2$ = 18 — all worked correctly.

2(iv) — MAIN Electron affinity; variation across a period and down a group — 2.5/3

Electron affinity defined (energy released when an electron is added to a neutral, gaseous atom to form a negative ion) ✓. Across a period: increases — size decreases, nuclear charge increases ✓. Down a group: decreases, with the correct primary reason (atomic size increases) ✓.

−0.5: she also wrote that down a group “nuclear charge **decreases**” — factually wrong. Nuclear charge *increases* down a group (more protons); electron affinity falls because atomic size and shielding grow, so the added electron sits farther from the nucleus.

2(v) — OR Formation of MgO by electron transfer — 3/3

Mg (2,8,2) has 2 valence electrons; O (2,6) needs 2. Mg loses 2e → Mg^{2+} (2,8, neon config); O gains those 2e → O^{2-} (2,8, neon config); the oppositely charged ions attract by a strong electrostatic force = ionic bond, forming MgO. All three marks present.

2(vi) — MAIN Define molar mass; molar mass of Na_2CO_3 — 3/3

Molar mass defined (mass in grams of one mole; numerically equal to the formula mass in g/mol) ✓. Na_2CO_3 = 2(23) + 12 + 3(16) = 46 + 12 + 48 = 106 g/mol ✓. Working and answer correct.

2(vii) — OR Balance Fe + H_2O (steam) → Fe_3O_4 + H_2; add state symbols — 2/3

Balancing perfect: 3Fe + $4H_2O$ → Fe_3O_4 + $4H_2$, checked atom-by-atom (Fe 3:3, H 8:8, O 4:4) — 2 marks.

−1: state symbols not added. The question explicitly says “add state symbols,” but the balanced equation carries no (s)/(l)/(g) — only the word “(steam)” beside H_2O on the copied line, which is a description, not a state symbol.

2(viii) — MAIN Oxidation state of the central atom in MnO_2, $HClO_4$, H_3PO_4 — 3/3

(a) Mn: x + 2(−2) = 0 → +4 ✓ (b) Cl: 1 + x + 4(−2) = 0 → +7 ✓ (c) P: 3(1) + x + 4(−2) = 0 → +5 ✓. All three with working.

2(ix) — OR **Why aluminium resists corrosion (oxide layer)** **3/3**

Aluminium is reactive but forms a thin, hard, continuous layer of Al_2O_3 on its surface; this layer is impervious and sticks tightly, sealing the metal underneath so no further air or moisture can reach it. Fully correct.

2(x) — OR **Define catalyst; industrial example; effect on rate and ΔH** **3/3**

Catalyst defined (increases the rate of a reaction without itself being consumed) ✓. Industrial example: Fe in the Haber process ($N_2 + 3H_2 \rightarrow 2NH_3$) ✓. (a) increases the rate by providing a lower-activation-energy path ✓ (b) does not change ΔH ✓.

2(xi) — MAIN **Physical vs chemical equilibrium; one example each** **3/3**

Physical equilibrium = balance in a reversible physical change, no new substance (water ⇌ water vapour in a closed bottle); chemical equilibrium = balance in a reversible chemical change, new substance formed ($N_2 + 3H_2 \rightarrow 2NH_3$). Difference plus both examples correct. (She labelled it "(or)" but wrote the main differentiate answer in full.)

Section C — Long Questions (17.5/20)

3(i) — MAIN **(a) Electronic config of C & S + group/period (3) | (b) atom with 12p/12e (2)** **5/5**

(a) Config defined ✓; Carbon $1s^22s^22p^2$, Group IV-A (14), Period 2 ✓; Sulphur $1s^22s^22p^63s^23p^4$, Group VI-A (16), Period 3 ✓ (sub-shell notation used — more detailed than the shells the key shows, still correct). (b) 12p/12e = Mg ($1s^22s^22p^63s^2$): (i) a metal, only 2 valence electrons it tends to lose ✓ (ii) forms Mg^{2+} ✓ (iii) valency 2 ✓. Full marks.

3(ii) — MAIN **(a) empirical/molecular formula; CH_2O, M=180 (3) | (b) molecules in 3.6 g H_2O (2)** **3/5**

(a) Empirical and molecular formula both defined ✓; CH_2O empirical mass = 30; n = 180/30 = 6; molecular formula = $(CH_2O)_6$ = $C_6H_{12}O_6$ (glucose) ✓ — full 3 marks.

> −2 (biggest single loss): part (b) — number of molecules in 3.6 g of water — was never attempted. The page runs straight from "$C_6H_{12}O_6$ (glucose)" into Q3(iii); no moles = 3.6/18 = 0.2 and no $\times 6.022\times10^{23}$ appears anywhere. She does the identical mole→molecules step correctly in 2(vi) and 3(iv), so this is a skipped sub-part, not a knowledge gap.

3(iii) — OR **(a) define + identify oxidizing/reducing agent in MnO_2+4HCl (3) | (b) balance Fe+Cl_2 (2)** **4.5/5**

Identification perfect: in $MnO_2 + 4HCl \rightarrow MnCl_2 + Cl_2 + 2H_2O$, Mn +4→+2 (reduced) = MnO_2 is the oxidizing agent ✓; Cl −1→0 (oxidized) = HCl is the reducing agent ✓. (b) $2Fe + 3Cl_2 \rightarrow 2FeCl_3$ ✓; Fe oxidized 0→+3 ✓.

> −0.5: the two definitions have the "causes another substance to…" clause reversed — she wrote an oxidizing agent "causes another substance to **gain** electrons" (should be *lose*) and a reducing agent "causes another to **lose**" (should be *gain*). The "is itself reduced / oxidised" halves are correct and she applied them correctly below, so only half a mark off.

3(iv) — OR **(a) ΔH for H_2+Br_2→2HBr (3) | (b) respiration & photosynthesis ΔH sign (2)** **5/5**

Bonds broken H–H 436 + Br–Br 193 = 629; bonds formed 2×(H–Br) = 2×366 = 732; ΔH = 629 − 732 = −103 kJ, so the reaction is exothermic ✓. (b)(i) respiration exothermic, ΔH negative ✓ (b)(ii) photosynthesis endothermic, ΔH positive ✓. Full marks. (Trivial: "−103 J" written once for kJ — not penalised.)

Overall

Strengths in this attempt

- MCQs 11/12 — every atomic-structure, bonding and stoichiometry MCQ correct; only the equilibrium-conditions item slipped.
- Stoichiometry rock-solid: molar mass of Na_2CO_3 (106), empirical→molecular formula ($C_6H_{12}O_6$), all three oxidation-state assignments, and the bond-energy ΔH (−103 kJ) all correct.
- Redox *application* spot-on — correctly identified the oxidizing and reducing agents and balanced $2Fe + 3Cl_2 \rightarrow 2FeCl_3$.
- MgO electron-transfer and the Al_2O_3 oxide-layer answers are complete and well-structured; catalyst and equilibrium answers full marks.

Areas to revise

- **Un-attempted sub-parts (−2, biggest loss):** 3(ii)(b) left blank. Same "half a long question skipped" pattern seen on Biology exam-08/09. Drill: before leaving a question, count how many parts it asks (a/b/i/ii) and tick each.
- **Oxidizing / reducing agent definitions (−0.5):** an oxidizing agent takes electrons *from* the other substance (oxidises it) and is itself reduced; a reducing agent gives electrons and is itself oxidised.
- **Electron affinity down a group (−0.5):** nuclear charge *increases* down a group; EA falls because of larger size and shielding.
- **State symbols (−1):** when told to "add state symbols," write (s)/(l)/(g) on every species in the balanced equation.
- **MCQ Q12:** a reversible reaction reaches equilibrium only in a *closed system*.

Chemistry Attempts — Comparison

Date	Exam	MCQ	Total	%	Grade
12 Apr 2026	exam-01-ch1-2	12/12	64/65	98.5%	A+
17 Apr 2026	exam-02-ch3-5	12/12	59.5/65	91.5%	A+
12 May 2026	exam-03-ch6	12/12	57.5/65	88.5%	A+
12 Jun 2026	exam-04-ch7	12/12	60/65	92.3%	A+
24 Jun 2026	exam-12-ch1-9	12/12	62.5/65	96.2%	A+
30 Jun 2026	exam-13-ch1-9	12/12	60.5/65	93.1%	A+
6 Jul 2026	**exam-14-ch1-9 (this exam)**	**11/12**	**60.0/65**	**92.3%**	**A+**

exam-14 (60.0/65, 92.3%) is Chemistry's third full-syllabus Ch1–9 paper, forming a tight cluster with exam-12 (96.2%) and exam-13 (93.1%). It carries Chemistry's first-ever MCQ error, so the 13-exam perfect-MCQ streak ends here. Even so it is another clean A+; the Chemistry subject average holds at 93.2% across 7 papers. The two recoverable habits — finishing every sub-part and adding state symbols — would have lifted this to 63/65.